



Interference Phenomenon with Concurrent Strength and High-Intensity Interval Training-Based Aerobic Training: An Updated Model

Felipe C. Vechin^{1,2} · Miguel S. Conceição¹ · Guilherme D. Telles¹ · Cleiton A. Libardi² · Carlos Ugrinowitsch¹

Accepted: 20 December 2020 / Published online: 6 January 2021

© The Author(s), under exclusive licence to Springer Nature Switzerland AG part of Springer Nature 2021

Abstract

Previous research has suggested that concurrent training (CT) may attenuate resistance training (RT)-induced gains in muscle strength and mass, i.e., the interference effect. In 2000, a seminal theoretical model indicated that the interference effect should occur when high-intensity interval training (HIIT) (repeated bouts at 95–100% of the aerobic power) and RT (multiple sets at ~10 repetition maximum; 10 RM) were performed in the same training routine. However, there was a paucity of data regarding the likelihood of other HIIT-based CT protocols to induce the interference effect at the time. Thus, based on current HIIT-based CT literature and HIIT nomenclature and framework, the present manuscript updates the theoretical model of the interference phenomenon previously proposed. We suggest that very intense HIIT protocols [i.e., resisted sprint training (RST), and sprint interval training (SIT)] can greatly minimize the odds of occurring the interference effect on muscle strength and mass. Thus, very intensive HIIT protocols should be implemented when performing CT to avoid the interference effect. Long and short HIIT-based CT protocols may induce the interference effect on muscle strength when HIIT bout is performed before RT with no rest interval between them.

Key Points

HIIT-based concurrent training minimizes the chance of the interference effect on muscle strength gains and muscle hypertrophy when repeated-sprint training (RST) and sprint interval training (SIT) HIIT models are performed.

HIIT-based concurrent training may induce the interference effect on muscle strength gains but not on muscle hypertrophy when long HIIT protocols are performed (e.g., long bout durations i.e., > 1 min.) before the RT protocol with no rest interval between them.

1 Introduction

Concurrent training (CT) is characterized by performing resistance training (RT) and aerobic training in the same training routine [1–8]. Due to the increase in muscle strength (i.e., maximum voluntary contraction) and mass (i.e., whole muscle or muscle fiber cross-sectional area—hypertrophy), and aerobic power (i.e., $\text{VO}_{2\text{max}}$), CT programs have been highly recommended to improve both sport performance and overall health [1, 9–17]. However, CT may attenuate muscle strength and mass gains compared to RT alone, a phenomenon termed as the ‘interference effect’ [17–24].

Since the publication of the first evidence of the interference effect [18], several experimental studies have attempted to determine the training-related variables contributing to its occurrence [1, 17, 19, 21–23]. However, the heterogeneity of experimental designs and training protocols among studies has produced equivocal findings [6]. Alternatively, David Docherty and Ben Sporer [5] proposed an elegant model based on specific characteristics of the RT and aerobic training that may induce the interference effect (see original study for a detailed depiction of the model [5]). The authors suggested that when aerobic training is performed at intensities close to the maximum aerobic power (95–100%

✉ Felipe C. Vechin
felipevechin@usp.br

¹ School of Physical Education and Sport, University of Sao Paulo, Av. Prof. Mello Moraes, 65, Cidade Universitária, Sao Paulo, SP 05508-030, Brazil

² MUSCULAB, Laboratory of Neuromuscular Adaptations To Resistance Training, Department of Physical Education, Federal University of Sao Carlos, Sao Carlos, Brazil

$\text{VO}_{2\text{max}}$) using high-intensity interval training (HIIT) protocols and RT is performed with more than 10 repetition maximum (RM), gains in muscle strength and mass are blunted compared to RT alone due to antagonistic training-induced adaptations. However, more intensive HIIT protocols (e.g., repeated sprint training—RST, and sprint interval training—SIT) were not considered in the original model.

In the past, HIIT was generically characterized by repeated maximal-to-supramaximal and short-to-long duration efforts (e.g., 7 s up to 5 min) with relief periods between them [5, 6]. Specifically, Docherty and Sporer [5] model mentioned that HIIT-based CT protocols were characterized by long exercise durations and relief periods (e.g., 3 min:3 min, equal work to rest ratio 1:1). Seeking to better organize HIIT prescription, Buchheit and Laursen [25, 26] proposed an empirical model using the intensity and duration of the efforts as control variables that mediate the training-induced adaptations and training features (e.g., number of efforts and relief period duration). The authors classified HIIT into four intensity prescription zones as follows: (1) long HIIT intervals (intensity: $\approx 90\text{--}110\%$ of $\text{VO}_{2\text{max}}$ velocity— $v\text{VO}_{2\text{max}}$; duration: > 60 s); (2) short HIIT intervals (intensity $\approx 110\text{--}130\%$ of $v\text{VO}_{2\text{max}}$; duration: < 60 s); (3) repeated-sprint training (RST—intensity: $\approx 140\text{--}170\%$ of $v\text{VO}_{2\text{max}}$ or $75\text{--}85\%$ of the maximal sprint speed; duration: 3 to 10 s); and (4) sprint interval training (SIT—intensity: $> 170\%$ of $v\text{VO}_{2\text{max}}$ or $> 85\%$ of the maximal sprint speed; duration: 30–45 s). Importantly, all the aforementioned HIIT prescription zones effectively enhance maximum aerobic power [25], but results regarding HIIT-induced neuromuscular adaptations are equivocal. Compared to long HIIT, high-speed HIIT protocols (i.e., RST and SIT) impose greater neuromuscular demand as assessed by higher muscle activation during efforts and blood lactate concentrations during effort and relief periods [26–30]. Also, the work to rest ratio (about 1:8) for RST and SIT protocols ensures that every effort is supramaximal and recovery periods are long enough to maintain the intensity of the efforts [25, 26]. As a result, one may suggest that skeletal muscle anaerobic performance-related adaptations (e.g., gains in muscle strength and mass) may be greater in RST and SIT than long and short HIIT protocols. However, this suggestion requires further scrutiny due to equivocal findings.

Thus far, not many studies have investigated the effect of HIIT on muscle strength and mass [31–36]. Two studies showed increases in muscle cross-sectional area (CSA) between 5 and 7% [35, 36] after short HIIT and RST protocols, while others have failed to do so. The equivocal results were possibly due to methodological issues, such as differences in training volume and duration, muscle mass assessment (i.e., whole muscle vs muscle fiber cross-sectional area), and small sample sizes [31, 33, 34, 37, 38]. Recently, additional studies have investigated the effect of HIIT-based

CT models on muscle strength and mass [3, 8, 11, 16, 19, 21, 39–47]. Interestingly, data suggest that HIIT-based CT protocols (i.e., RST and SIT) impair neither muscle strength nor muscle mass gains compared to RT alone [11, 40–44]. These results are exciting due to the potential practical application of those CT protocols in sport and health contexts. Considering the recent framework used to prescribe HIIT protocols (i.e., long and short HIIT, RST and SIT) [25, 26], their respective neuromuscular adaptations and, thus, the chance of producing the interference effect, the purpose of this paper was to present an updated version of the CT interference phenomenon model proposed by Docherty and Sporer [5]. The literature was searched using PubMed, Web of Science and Google Scholar up to August 2020.

2 Interference in Muscle Strength Gains in HIIT-Based CT Protocols

The original Docherty and Sporer model [5] proposes that low-intensity high-volume aerobic training and high-intensity low-volume RT produce mainly central adaptations (cardiovascular and nervous system, respectively). Due to those specific characteristics, they are at opposing extremes of the CT adaptations continuum and should not produce the interference effect on muscle strength [5]. Conversely, high-intensity interval training (HIIT) and RT usually implemented in CT protocols are not at the extremes of the continuum, producing not only central adaptations but also peripheral ones (i.e., skeletal muscle). The peripheral adaptations induced by HIIT and RT can be in opposite directions, which has been deemed to be a leading cause of the interference effect. According to the original interference phenomenon model [5], long HIIT can induce the interference effect. As the other three HIIT prescription zones [25] impose distinct demands on skeletal muscles compared to long HIIT [26], the occurrence of the interference effect when using them requires further scrutiny. For instance, RST and SIT models have higher glycolytic demand than long and short HIIT models [26]. Additionally, RST and SIT include maximum or near maximum sprints, which enhance muscle activation and force production compared to long and short HIIT [29, 30]. Similarly, moderate-to-high intensity RT (i.e., > 10 RM) requires high muscle activation, force production and rate of energy production (i.e., glycolytic demand) [48–52], factors associated with increases in muscle strength [53, 54]. Taken together, it is reasonable to suggest that RST- and SIT-based CT protocols should not produce antagonistic adaptations and thus not impair muscle strength gains [55].

Sabag and colleagues [11] published a meta-analysis showing a marginal interference effect on muscle strength of HIIT-based CT (effect size = -0.248 ; 95% confidence interval -0.495 to -0.001) compared to RT alone.

However, the authors did not use the prescription zones as a moderator variable in their statistical model to determine the likelihood of each zone producing the interference effect. Thus, we critically analyzed the 12 studies included in Sabag and colleagues' [11] work, and two additional studies retrieved in the search process [47, 56]. Only three studies [19, 44, 56] showed reduced gains in muscle strength when comparing HIIT-based CT protocols with RT alone. Fyfe et al. [19] and Robineau et al. [44] used long and short HIIT-based CT protocols, respectively, with no rest interval between the HIIT and RT bouts. Similarly, Chtara et al. [56] also used a long HIIT-based CT protocol with a 15 min rest interval between bouts. The rest interval between exercise bouts has been considered as an important CT-related variable [23] that may favor the occurrence of the interference effect. Accordingly, Robineau et al. [43] compared the effect of 0 h, 6 h and 24 h interval between the RT and short HIIT bouts on muscle strength gains, the 6 h interval attenuated the interference effect and the 24 h interval eliminated it. Other training-related variables, such as exercise mode (i.e., running or cycling), exercise order (HIIT-RT or RT-HIIT), previous training experience, intervention duration, between-individual variability and high variability of the prescribed training protocols, can modulate the occurrence of the interference effect [1, 6, 11, 17, 23, 44, 47, 56]. However, designing studies able to control for those confounding variables is challenging and efforts should be made to standardize training protocols. Thus, current evidence indicates that Docherty and Sporer's interference model [5] should be updated, as more intensive HIIT protocols (RST and SIT) are unlikely to hamper muscle strength gains (Fig. 1) while long and short HIIT protocols could blunt them (Fig. 1).

Even though most of the studies have determined the effect of HIIT-based CT protocols on muscle strength gains, improvements in muscle power (i.e., countermovement jump [CMJ] height) could also be considered in the

interference model [3, 19, 39, 43, 44, 47, 56], due to its importance to sport performance and daily life activities. Three studies demonstrated lesser improvements in CMJ height when using long and short HIIT-based CT protocols [19, 47, 56], while studies using RST- and SIT-based CT protocols did not show the interference effect [3, 39, 43, 44]. Similar to muscle strength, adding a rest interval between the HIIT and RT bouts (no interval vs. more than 6 h of interval) also seems to decrease the odds of occurrence of the interference effect [44]. However, most of the studies used traditional RT protocols and, thus, there is a paucity of data on the occurrence of the interference effect when performing more power-oriented RT protocols and exercises (e.g., velocity-based training and bench throw, respectively).

3 Muscle Hypertrophy Interference Model Based on HIIT Evidence

Current exercise prescription guidelines indicate that RT-induced gains in muscle mass are obtained by performing multiple sets of 3–12 RM (70–100% 1-RM) (depending on training status) [53, 54]. Recently, mounting evidence suggests that low-intensity RT (20–50% 1-RM), performed close to concentric failure, produces similar gains in muscle mass to high-intensity RT [57, 58]. Thus, effort duration, rest interval, and skeletal muscle metabolic demand seem to be similar between hypertrophy-oriented RT and all four HIIT prescription zones, indicating that the gains in muscle mass should not be affected by HIIT-based CT protocols. In fact, there is evidence that specific HIIT prescription zones can increase the mass of the exercised muscles.

Accordingly, Linossier et al. [35] submitted healthy and physically active young men to an RST (5 s sprint, 55 s rest; two sets of 15 sprints) protocol for nine weeks, four

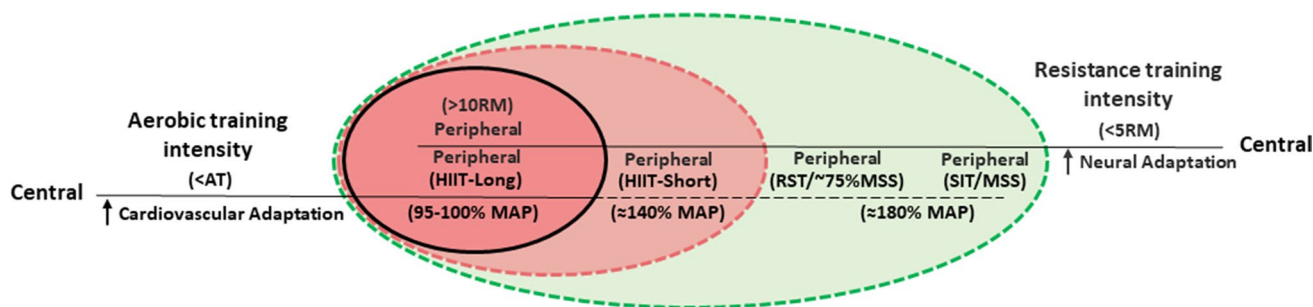


Fig. 1 Updated concurrent training interference effect model for muscle strength. *AT* aerobic threshold, *MAP* maximal aerobic power, *RM* repetition maximum, *HIIT* high-intensity interval training, *MSS* maximal sprinting speed, *RST* repeated-sprint training, *SIT* sprint interval training. The pink area inside the solid black ellipse indicates the

original interference zone; the light pink area inside the dashed red ellipse indicates a slight interference zone; and the green area inside the dashed green ellipse indicates a non-interference zone (adapted from Docherty and Sporer [5])

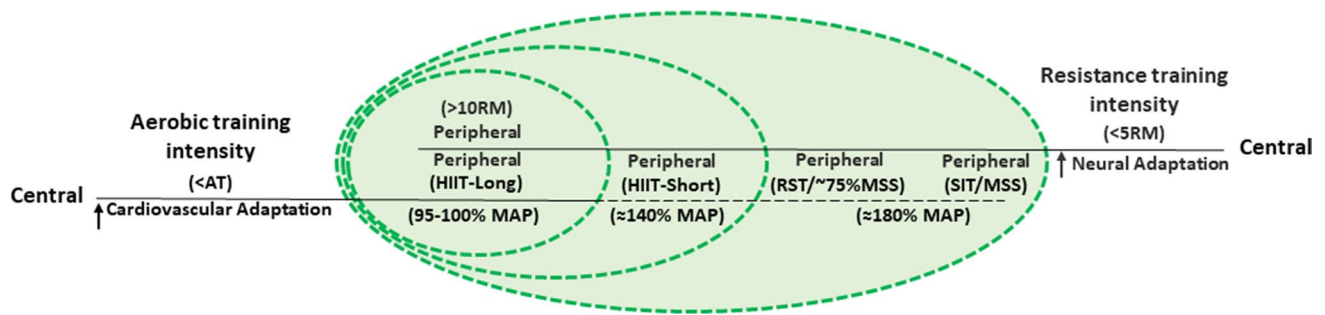


Fig. 2 Updated concurrent training interference effect model for muscle mass. *AT* aerobic threshold, *MAP* maximal aerobic power, *RM* repetition maximum, *HIIT* high-intensity interval training, *MSS* maxi-

mal sprinting speed, *RST* repeated-sprint training, *SIT* sprint interval training. The green area inside the dashed green ellipse lines indicate non-interference zones (adapted from Docherty and Sporer [5])

times per week. Subjects increased quadriceps, vastus lateralis, and vastus lateralis type II fiber cross-sectional area (CSA) by 5%, 6%, and 45%, respectively. Similarly, Osawa et al. [36] reported an 11% increase in quadriceps CSA after a 16 week short HIIT (60 s sprint, 60 s rest; 8–12 sprints; > 90% $\text{VO}_{2\text{peak}}$) protocol, twice a week. In fact, not only RST and short HIIT but also long HIIT seems to increase muscle mass. Estes et al. [59] reported an 11% increase in vastus lateralis CSA in recreationally active young participants after 10 weeks of long HIIT (24 sessions; 4 min effort at 90–95% HRmax and 3 min relief at 70% HRmax). On the other hand, others have failed to demonstrate RST- and SIT-induced gains in muscle fiber CSA [31, 33, 37, 38]. Thus, there is some evidence that HIIT can induce muscle hypertrophy, but additional studies are required to determine its effects at the muscle fiber level.

Other studies have compared the gains in muscle mass between HIIT-based CT protocols and RT alone protocols [8, 19, 22, 39–42]. As evidence suggests that HIIT protocols can increase [35, 36, 59] or have no effect [31, 33, 37, 38] on muscle mass, it is reasonable to suggest that HIIT-based CT models could preclude the occurrence of the interference effect on muscle mass. One may even suggest an additive effect, as HIIT can increase muscle mass per se; however there is no empirical evidence of this additive effect [40–42]. There are studies comparing long HIIT, short HIIT, RST- and SIT-based CT protocols with RT alone on muscle mass gains. Independently of HIIT prescription zone, no interference effect was observed on muscle mass [22, 39, 40, 42]. Considering the HIIT-based CT literature, only Fyfe et al. [19] showed a possible interference effect on muscle mass gains when comparing a RT protocol with a long HIIT-based CT one ($4.1 \pm 2.0\%$ and $1.8 \pm 1.6\%$, respectively). However, change in lower-body lean mass was used as a proxy of muscle mass gain, which may have biased the findings. Changes from pre- to

post-training were small and represented the sum of the changes in mass of all lower body muscles (i.e., false positive). Furthermore, training sessions followed a HIIT-RT order, which may favor the interference effect [23].

Taken together, Sabag and colleagues' meta-analysis [11], Murach and Bagley [17], and Lee et al. [47] do not suggest the occurrence of the interference effect on muscle mass when performing HIIT-based CT protocols (Fig. 2). Evidently, additional studies are necessary to test if differences in total training volume, exercise order, training frequency, training experience, and exercise mode may induce the interference effect when performing HIIT-based CT. For instance, Murach and Bagley [17] suggested that cycling HIIT-based CT protocols might produce the aforementioned additive effect on muscle hypertrophy when compared to running HIIT-based CT protocols. On the other hand, Sabag and colleagues [11] did not find differences in muscle mass gains between running and cycling HIIT-based CT protocols. Considering that most of the studies investigating different HIIT-based CT protocols showed no interference effect on muscle mass, it is reasonable to suggest that different combinations of HIIT and RT protocols may be used without compromising muscle mass gains (Fig. 2).

4 Conclusion

More intense HIIT protocols, such as RST and SIT, when combined with RT can promote similar gains in muscle strength, mass, and power to RT alone. It is noteworthy that if HIIT induces the interference effect, it affects muscle strength and CMJ height gains mainly when using long and short HIIT protocols. Finally, performing a long or short HIIT bout after RT with a proper rest interval between them (> 6 h) may decrease the probability of occurrence of the interference effect on muscle strength.

4.1 Practical Implications

The evidence presented herein suggests that different HIIT protocols may be used when performing CT. Thus, the selection of specific HIIT protocols should be based on personal preferences and training goals (e.g., different sport's needs, health, or physical fitness promotion). If the interference effect has to be avoided at all costs, evidence suggests that RST and SIT should be preferred as the likelihood of occurrence of the interference effect on muscle strength, mass and power is then low.

Declarations

Authorship contributions FC conceived the idea, wrote the first draft, worked on all drafts and formatted the manuscript for submission. MSC, GDT CAL and CU helped develop the main idea and draft the paper. All authors read and approved the final version of the manuscript.

Funding The present article was supported by the São Paulo Research Foundation (FAPESP) (Grant no. #2018/12150-0 to C.A.L.), and National Council for Scientific and Technological Development (Grant no. #302801/2018-9 to C.A.L. and Grant no. #303085/2015-0 to C.U.).

Conflicts of interest/Competing interests Felipe Cassaro Vechin, Miguel S. Conceição, Guilherme D. Telles, Cleiton A. Libardi and Carlos Ugrinowitsch declare no conflict of interest relevant to the content of this article.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication The manuscript has been read and approved for submission by all the named authors.

Availability of data and material Not applicable.

Code availability Not applicable.

References

- Coffey VG, Hawley JA. Concurrent exercise training: do opposites distract? *J Physiol*. 2017;595(9):2883–96.
- Hakkinen K, Alen M, Kraemer WJ, Gorostiaga E, Izquierdo M, Rusko H, et al. Neuromuscular adaptations during concurrent strength and endurance training versus strength training. *Eur J Appl Physiol*. 2003;89(1):42–52.
- Balabinis CP, Psarakis CH, Moukas M, Vassiliou MP, Behrakis PK. Early phase changes by concurrent endurance and strength training. *J Strength Cond Res*. 2003;17(2):393–401.
- McCarthy JP, Pozniak MA, Agre JC. Neuromuscular adaptations to concurrent strength and endurance training. *Med Sci Sports Exerc*. 2002;34(3):511–9.
- Docherty D, Sporer B. A proposed model for examining the interference phenomenon between concurrent aerobic and strength training. *Sports Med*. 2000;30(6):385–94.
- Leveritt M, Abernethy PJ, Barry BK, Logan PA. Concurrent strength and endurance training. *A Rev Sports Med*. 1999;28(6):413–27.
- Sale DG, MacDougall JD, Jacobs I, Garner S. Interaction between concurrent strength and endurance training. *J Appl Physiol*. 1990;68(1):260–70.
- Sale DG, Jacobs I, MacDougall JD, Garner S. Comparison of two regimens of concurrent strength and endurance training. *Med Sci Sports Exerc*. 1990;22(3):348–56.
- Sousa AC, Neiva HP, Izquierdo M, Cadore EL, Alves AR, Marinho DA. Concurrent training and detraining: brief review on the effect of exercise intensities. *Int J Sports Med*. 2019;40(12):747–55. <https://doi.org/10.1055/a-0975-9471>.
- Berryman N, Mujika I, Bosquet L. Concurrent training for sports performance: the 2 sides of the medal. *Int J Sports Physiol Perform*. 2019;14(3):279–85.
- Sabag A, Najafi A, Michael S, Esgin T, Halaki M, Hackett D. The compatibility of concurrent high intensity interval training and resistance training for muscular strength and hypertrophy: a systematic review and meta-analysis. *J Sports Sci*. 2018;36(21):2472–83.
- Cadore EL, Pinto RS, Teodoro JL, da Silva LXN, Menger E, Alberton CL, et al. Cardiorespiratory adaptations in elderly men following different concurrent training regimes. *J Nutr Health Aging*. 2018;22(4):483–90.
- De Luca V, Minganti C, Borriore P, Grazioli E, Cerulli C, Guerra E, et al. Effects of concurrent aerobic and strength training on breast cancer survivors: a pilot study. *Public health*. 2016;136:126–32.
- Botonis PG, Toubekis AG, Platanou TI. Concurrent strength and interval endurance training in elite water polo players. *J Strength Cond Res*. 2016;30(1):126–33.
- Libardi CA, Chacon-Mikahil MP, Cavaglieri CR, Tricoli V, Roschel H, Vechin FC, et al. Effect of concurrent training with blood flow restriction in the elderly. *Int J Sports Med*. 2015;36(5):395–9.
- Mijwel S, Backman M, Bolam KA, Olofsson E, Norrbom J, Bergh J, et al. Highly favorable physiological responses to concurrent resistance and high-intensity interval training during chemotherapy: the OptiTrain breast cancer trial. *Breast Cancer Res Treat*. 2018;169(1):93–103.
- Murach KA, Bagley JR. Skeletal muscle hypertrophy with concurrent exercise training: contrary evidence for an interference effect. *Sports Med*. 2016;46(8):1029–39.
- Hickson RC. Interference of strength development by simultaneously training for strength and endurance. *Eur J Appl Physiol Occup Physiol*. 1980;45(2–3):255–63.
- Fyfe JJ, Bartlett JD, Hanson ED, Stepto NK, Bishop DJ. Endurance training intensity does not mediate interference to maximal lower-body strength gain during short-term concurrent training. *Front Physiol*. 2016;7:487.
- Fyfe JJ, Bishop DJ, Stepto NK. Interference between concurrent resistance and endurance exercise: molecular bases and the role of individual training variables. *Sports Med*. 2014;44(6):743–62.
- de Souza EO, Tricoli V, Aoki MS, Roschel H, Brum PC, Bacurau AV, et al. Effects of concurrent strength and endurance training on genes related to myostatin signaling pathway and muscle fiber responses. *J Strength Cond Res*. 2014;28(11):3215–23.
- de Souza EO, Tricoli V, Roschel H, Brum PC, Bacurau AV, Ferreira JC, et al. Molecular adaptations to concurrent training. *Int J Sports Med*. 2013;34(3):207–13.
- Wilson JM, Marin PJ, Rhea MR, Wilson SM, Loenneke JP, Anderson JC. Concurrent training: a meta-analysis examining interference of aerobic and resistance exercises. *J Strength Cond Res*. 2012;26(8):2293–307.

24. Nader GA. Concurrent strength and endurance training: from molecules to man. *Med Sci Sports Exerc.* 2006;38(11):1965–70.
25. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle: part I: cardiopulmonary emphasis. *Sports Med.* 2013;43(5):313–38.
26. Buchheit M, Laursen PB. High-intensity interval training, solutions to the programming puzzle. Part II: anaerobic energy, neuromuscular load and practical applications. *Sports Med.* 2013;43(10):927–54.
27. Girard O, Mendez-Villanueva A, Bishop D. Repeated-sprint ability—part I: factors contributing to fatigue. *Sports Med.* 2011;41(8):673–94.
28. Bishop D, Girard O, Mendez-Villanueva A. Repeated-sprint ability—part II: recommendations for training. *Sports Med.* 2011;41(9):741–56.
29. van den Tillaar R, Solheim JAB, Bencke J. Comparison of hamstring muscle activation during high-speed running and various hamstring strengthening exercises. *Int J Sports Phys Therapy.* 2017;12(5):718–27.
30. Higashihara A, Ono T, Kubota J, Okuwaki T, Fukubayashi T. Functional differences in the activity of the hamstring muscles with increasing running speed. *J Sports Sci.* 2010;28(10):1085–92.
31. Mueller SM, Aguayo D, Zuercher M, Fleischmann O, Boutellier U, Auer M, et al. High-intensity interval training with vibration as rest intervals attenuates fiber atrophy and prevents decreases in anaerobic performance. *PLoS ONE.* 2015;10(2):e0116764.
32. De Smet S, Van Thienen R, Deldicque L, James R, Sale C, Bishop DJ, et al. Nitrate intake promotes shift in muscle fiber type composition during sprint interval training in hypoxia. *Front Physiol.* 2016;7:233.
33. Kohn TA, Essen-Gustavsson B, Myburgh KH. Specific muscle adaptations in type II fibers after high-intensity interval training of well-trained runners. *Scand J Med Sci Sports.* 2011;21(6):765–72.
34. Linossier MT, Denis C, Dormois D, Geyssant A, Lacour JR. Ergometric and metabolic adaptation to a 5-s sprint training programme. *Eur J Appl Physiol Occup Physiol.* 1993;67(5):408–14.
35. Linossier MT, Dormois D, Geyssant A, Denis C. Performance and fibre characteristics of human skeletal muscle during short sprint training and detraining on a cycle ergometer. *Eur J Appl Physiol Occup Physiol.* 1997;75(6):491–8.
36. Osawa Y, Azuma K, Tabata S, Katsukawa F, Ishida H, Oguma Y, et al. Effects of 16-week high-intensity interval training using upper and lower body ergometers on aerobic fitness and morphological changes in healthy men: a preliminary study. *Open Access J Sports Med.* 2014;5:257–65.
37. Joanisse S, Gillen JB, Bellamy LM, McKay BR, Tarnopolsky MA, Gibala MJ, et al. Evidence for the contribution of muscle stem cells to nonhypertrophic skeletal muscle remodeling in humans. *FASEB J.* 2013;27(11):4596–605.
38. Joanisse S, McKay BR, Nederveen JP, Scribbans TD, Gurd BJ, Gillen JB, et al. Satellite cell activity, without expansion, after nonhypertrophic stimuli. *Am J Physiol Regul Integr Comp Physiol.* 2015;309(9):R1101–11.
39. Tsitkanou S, Spengos K, Stasinaki AN, Zaras N, Bogdanis G, Papadimas G, et al. Effects of high-intensity interval cycling performed after resistance training on muscle strength and hypertrophy. *Scand J Med Sci Sports.* 2017;27(11):1317–27.
40. Laird RHT, Elmer DJ, Barberio MD, Salom LP, Lee KA, Pascoe DD. Evaluation of performance improvements after either resistance training or sprint interval-based concurrent training. *J Strength Cond Res.* 2016;30(11):3057–65.
41. Kikuchi N, Yoshida S, Okuyama M, Nakazato K. The effect of high-intensity interval cycling sprints subsequent to arm-curl exercise on upper-body muscle strength and hypertrophy. *J Strength Cond Res.* 2016;30(8):2318–23.
42. Cantrell GS, Schilling BK, Paquette MR, Murlasits Z. Maximal strength, power, and aerobic endurance adaptations to concurrent strength and sprint interval training. *Eur J Appl Physiol.* 2014;114(4):763–71.
43. Robineau J, Lacombe M, Piscione J, Bigard X, Babault N. Concurrent training in rugby sevens: effects of high-intensity interval exercises. *Int J Sports Physiol Perform.* 2017;12(3):336–44.
44. Robineau J, Babault N, Piscione J, Lacombe M, Bigard AX. Specific training effects of concurrent aerobic and strength exercises depend on recovery duration. *J Strength Cond Res.* 2016;30(3):672–83.
45. Ross RE, Ratamess NA, Hoffman JR, Faigenbaum AD, Kang J, Chilakos A. The effects of treadmill sprint training and resistance training on maximal running velocity and power. *J Strength Cond Res.* 2009;23(2):385–94.
46. Silva RF, Cadore EL, Kothe G, Guedes M, Alberton CL, Pinto SS, et al. Concurrent training with different aerobic exercises. *Int J Sports Med.* 2012;33(8):627–34.
47. Lee MJ, Ballantyne JK, Chagolla J, Hopkins WG, Fyfe JJ, Phillips SM, et al. Order of same-day concurrent training influences some indices of power development, but not strength, lean mass, or aerobic fitness in healthy, moderately-active men after 9 weeks of training. *PLoS ONE.* 2020;15(5):e0233134.
48. Looney DP, Kraemer WJ, Joseph MF, Comstock BA, Denegar CR, Flanagan SD, et al. Electromyographical and perceptual responses to different resistance intensities in a squat protocol: does performing sets to failure with light loads produce the same activity? *J Strength Cond Res.* 2016;30(3):792–9.
49. Gonzalez AM, Ghigiarelli JJ, Sell KM, Shone EW, Kelly CF, Mangine GT. Muscle activation during resistance exercise at 70% and 90% 1-repetition maximum in resistance-trained men. *Muscle Nerve.* 2017;56(3):505–9.
50. McCauley GO, McBride JM, Cormie P, Hudson MB, Nuzzo JL, Quindry JC, et al. Acute hormonal and neuromuscular responses to hypertrophy, strength and power type resistance exercise. *Eur J Appl Physiol.* 2009;105(5):695–704.
51. Gonzalez AM, Hoffman JR, Townsend JR, Jajtner AR, Boone CH, Beyer KS, et al. Intramuscular anabolic signaling and endocrine response following high volume and high intensity resistance exercise protocols in trained men. *Physiol Rep.* 2015;3(7):e12466.
52. Smilios I, Pilianidis T, Karamouzis M, Tokmakidis SP. Hormonal responses after various resistance exercise protocols. *Med Sci Sports Exerc.* 2003;35(4):644–54.
53. Campos GE, Luecke TJ, Wendeln HK, Toma K, Hagerman FC, Murray TF, et al. Muscular adaptations in response to three different resistance-training regimens: specificity of repetition maximum training zones. *Eur J Appl Physiol.* 2002;88(1–2):50–60.
54. Ratamess NA, Alvar BA, Evetoch TK, Housh TJ, Kibler WB, Kraemer WJ, et al. American College of Sports Medicine position stand. Progression models in resistance training for healthy adults. *Med Sci Sports Exerc.* 2009;41(3):687–708.
55. Sale DG. Neural adaptations to resistance training. *Med Sci Sports Exerc.* 1988;20(5(Supplement)):S135–45.
56. Chtara M, Chaouachi A, Levin GT, Chaouachi M, Chamari K, Amri M, et al. Effect of concurrent endurance and circuit resistance training sequence on muscular strength and power development. *J Strength Cond Res.* 2008;22(4):1037–45.
57. Nobrega SR, Ugrinowitsch C, Pintanel L, Barcelos C, Libardi CA. Effect of resistance training to muscle failure vs. Volitional

- interruption at high- and low-intensities on muscle mass and strength. *J Strength Cond Res.* 2018;32(1):162–9.
58. Schoenfeld BJ, Grgic J, Ogborn D, Krieger JW. Strength and hypertrophy adaptations between low- versus high-load resistance training: a systematic review and meta-analysis. *J Strength Cond Res.* 2017;31(12):3508–23. <https://doi.org/10.1519/JSC.0000000000002200>.
59. Estes RR, Malinowski A, Piacentini M, Thrush D, Salley E, Losey C, et al. The effect of high intensity interval run training on cross-sectional area of the vastus lateralis in untrained college students. *Int J Exerc Sci.* 2017;10(1):137–45.